

Jerry Xu

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PROFESSIONAL SUMMARY

Senior Software Engineer specializing in large-scale distributed systems and control plane architecture. Leads high-impact infrastructure initiatives on Azure's SDN platform, improving reliability, scalability, and performance across millions of resources. Proven ability to drive cross-organizational designs and deliver production-critical systems at global scale.

PROFESSIONAL EXPERIENCE

Microsoft | Azure SDN Control Plane

2019 – Present

Software Engineer → Software Engineer II (2020) → Senior Software Engineer (2022)

- Drove the integration of regional network management into Azure's new SDN control plane, implementing pub/sub-based goal-state publishing for VM, NIC, and network policy resources via Protobuf contracts. Part of a multi-org effort that reduced VM lifecycle latency by 90%.
- Led cross-organizational design with Azure Compute to redesign the VM deletion API, eliminating resource-conflict incidents caused by premature SDN release and enabling migration of the VM lifecycle to a modern IaaS architecture.
- Redesigned goal-state distribution to route publish calls to VM home availability zones, reducing cross-AZ traffic by 66–75% and improving resilience during zone-level pubsub outages.
- Built a proactive error-triage workflow to surface low-frequency failures below alerting thresholds, evolving it from manual review to AI-driven classification and uncovering previously undetected production issues.
- Onboarded a new 6-person engineering team in India through structured weekly sessions, code review, and mentoring, enabling independent contributions within two months. The team has since grown to 15+ engineers.

Capital One Canada | Software Developer

2018

- Designed and prototyped real-time event-driven data streams using Apache Kafka for financial ledger updates, architected for zero data loss across ~2M Canadian cardholders.

IBM | Software Developer

2017 – 2018

- Built and maintained robust backend services in Java/C++, focusing on enterprise-scale data processing, storage reliability, and security practices including OWASP principles and GDPR compliance.

EDUCATION

McMaster University | Bachelor of Science in Computer Science, Minor in Business

2017

Dean's Honor List

TECHNICAL SKILLS

Languages: C#, Java, Python

Backend & Distributed Systems: API design, system design, scalability, microservices, REST APIs, gRPC, Pub/Sub, Protobuf, Apache Kafka, event-driven architecture, concurrent programming, multithreading

Cloud & Infrastructure: Azure, AWS, Service Fabric, SDN

Testing & Quality: unit testing, integration testing, TDD, code review

Engineering Practices: CI/CD, design documents, on-call, incident response, mentoring, technical leadership, cross-functional collaboration

Tools: Git, GitHub